

미적분 Black Half Test 05. 정답 및 해설

$$\begin{aligned}
 1) \quad & \int_a^b \frac{\cosh x}{\sinh^2 x + \sinh^4 x} dx \\
 &= \int_1^{\sqrt{3}} \frac{1}{t^2 + t^4} dt \\
 &= \int_1^{\sqrt{3}} \frac{1}{t^2(t^2 + 1)} dt \\
 &= \int_1^{\sqrt{3}} \left(\frac{1}{t^2} - \frac{1}{t^2 + 1} \right) dt \\
 &= \left[-\frac{1}{t} - \tan^{-1} t \right]_1^{\sqrt{3}} = 1 - \frac{1}{\sqrt{3}} - \frac{\pi}{12}
 \end{aligned}$$

정답 : ①

$$\begin{aligned}
 2) \quad & (\sec^{-1} x)' = \frac{1}{|x| \sqrt{x^2 - 1}} \text{ 이므로} \\
 & \int_{\sqrt{2}}^2 \frac{\sec^2(\sec^{-1} x)}{x \sqrt{x^2 - 1}} dx \\
 &= [\tan(\sec^{-1} x)]_{\sqrt{2}}^2 \\
 &= \tan(\sec^{-1} 2) - \tan(\sec^{-1} \sqrt{2}) \\
 &= \tan\left(\frac{\pi}{3}\right) - \tan\left(\frac{\pi}{4}\right) \\
 &= \sqrt{3} - 1
 \end{aligned}$$

정답 : ③

$$\begin{aligned}
 3) \quad & \int_1^{\sqrt{3}} \frac{\cot^{-1} x}{x^2} dx \\
 &= \left[-\frac{1}{x} \cot^{-1} x \right]_1^{\sqrt{3}} - \int_1^{\sqrt{3}} \left(-\frac{1}{x} \cdot \frac{-1}{1+x^2} \right) dx \quad (\because \text{부분적분}) \\
 &= -\frac{1}{\sqrt{3}} \cot^{-1} \sqrt{3} + \cot^{-1} 1 - \int_1^{\sqrt{3}} \frac{1}{x(1+x^2)} dx \\
 &= -\frac{\sqrt{3}}{3} \times \frac{\pi}{6} + \frac{\pi}{4} - \int_1^{\sqrt{3}} \frac{1}{x} - \frac{x}{1+x^2} dx \\
 &= -\frac{\sqrt{3}}{3} \times \frac{\pi}{6} + \frac{\pi}{4} - \left[\ln x - \frac{1}{2} \ln(1+x^2) \right]_1^{\sqrt{3}} \\
 &= -\frac{\sqrt{3}\pi}{18} + \frac{\pi}{4} - \left(\ln \sqrt{3} - \frac{1}{2} \ln(4) + \frac{1}{2} \ln(2) \right) \\
 &= -\frac{\sqrt{3}\pi}{18} + \frac{\pi}{4} + \frac{1}{2} \ln\left(\frac{2}{3}\right)
 \end{aligned}$$

정답 : ④

$$\begin{aligned}
 4) \quad & \sqrt[6]{x} = t, \quad \sqrt{x} = t^3, \quad \sqrt[3]{x} = t^2, \quad x = t^6, \quad dx = 6t^5 dt \text{ 이므로} \\
 & \int \frac{1}{\sqrt{x} - \sqrt[3]{x}} dx \\
 &= \int \frac{1}{t^3 - t^2} 6t^5 dt \\
 &= 6 \int \frac{t^3}{t-1} dt \\
 &= 6 \int (t^2 + t + 1 + \frac{1}{t-1}) dt \\
 &= 6 \left(\frac{1}{3} t^3 + \frac{1}{2} t^2 + t + \ln|t-1| \right) + C \\
 &= 2t^3 + 3t^2 + 6t + 6 \ln|t-1| + C \\
 &= 2\sqrt{x} + 3\sqrt[3]{x} + 6\sqrt[6]{x} + 6 \ln|\sqrt[6]{x}-1| + C \\
 \therefore a=2, b=3, c=6, d=6, \quad a+b+c+d=17
 \end{aligned}$$

정답 : ①

$$\begin{aligned}
 5) \quad & \int_0^1 (2x+1)^n dx \\
 &= \frac{1}{2} \frac{1}{n+1} [(2x+1)^{n+1}]_0^1 \\
 &= \frac{1}{2(n+1)} (3^{n+1} - 1) \\
 \Rightarrow \quad & \lim_{n \rightarrow \infty} \left\{ \int_0^1 (2x+1)^n dx \right\}^{\frac{1}{n}} \\
 &= \lim_{n \rightarrow \infty} \left\{ \frac{1}{2(n+1)} (3^{n+1} - 1) \right\}^{\frac{1}{n}} \\
 &= \lim_{n \rightarrow \infty} \frac{(3^{n+1} - 1)^{\frac{1}{n}}}{2^{\frac{1}{n}} (n+1)^{\frac{1}{n}}} = \frac{3}{1 \times 1} = 3
 \end{aligned}$$

$$1) \lim_{n \rightarrow \infty} (1+n)^{\frac{1}{n}} = \lim_{n \rightarrow \infty} e^{\ln(1+n)^{\frac{1}{n}}} = \lim_{n \rightarrow \infty} e^{\frac{\ln(1+n)}{n}} = e^0 = 1$$

$$2) \lim_{n \rightarrow \infty} 2^{\frac{1}{n}} = 2^0 = 1$$

$$\begin{aligned}
 3) \quad & \lim_{n \rightarrow \infty} (3^{n+1} - 1)^{\frac{1}{n}} = \lim_{n \rightarrow \infty} e^{\ln(3^{n+1} - 1)^{\frac{1}{n}}} \\
 &= \lim_{n \rightarrow \infty} e^{\frac{\ln(3^{n+1} - 1)}{n}} \\
 &= \lim_{n \rightarrow \infty} e^{\frac{3^{n+1} \ln 3}{3^{n+1} - 1}} = e^{\ln 3} = 3
 \end{aligned}$$

정답 : ③

$$\begin{aligned}
 6) \quad & f(x+y) = f(x) + f(y) + xy \text{ 에서 양변을 } x \text{ 에 대해 편미분하면} \\
 & f'(x+y) = f'(x) + y \text{ 이고 } x=0 \text{ 을 대입하면} \\
 & f'(0) = -2 \text{ 이므로 } f'(y) = -2 + y \text{ 이다.} \\
 & f'(x) = -2 + x \text{ 에서 적분하면 } f(x) = -2x + \frac{1}{2}x^2 + c \text{ 이고} \\
 & f(0) = 0 \text{ 이므로 } f(x) = -2x + \frac{1}{2}x^2 \text{ 이다.}
 \end{aligned}$$

$$\int_0^1 -2x + \frac{1}{2}x^2 dx = \left[-x^2 + \frac{1}{6}x^3 \right]_0^1 = -1 + \frac{1}{6} = -\frac{5}{6}$$

정답 : ②

$$\begin{aligned}
 7) \quad & f(\sqrt{2}) = 0 \text{ 이고} \\
 & \text{미적분학의 기본정리에 의하여 } f'(x) = \sinh^{-1} x \text{ 이다.}
 \end{aligned}$$

$$\text{따라서 } f''(x) = \frac{1}{\sqrt{1+x^2}} \text{ 이므로}$$

$$f'(\sqrt{2}) = \sinh^{-1}(\sqrt{2}) = \ln(\sqrt{2} + \sqrt{3}) \text{ 이고}$$

$$f''(\sqrt{2}) = \frac{1}{\sqrt{3}} \text{ 이다.}$$

$$\text{따라서 } g''(0) = -\frac{f''(\sqrt{2})}{(f'(\sqrt{2}))^3} = -\frac{1}{\sqrt{3}(\ln(\sqrt{2} + \sqrt{3}))^3} \text{ 이다.}$$

정답 : ②

$$\begin{aligned}
 8) \quad & \lim_{x \rightarrow \infty} \frac{\int_0^x \sqrt{1+\cos t} dt}{x} \quad (\because \text{로피탈 정리}) \\
 &= \lim_{x \rightarrow \infty} \sqrt{1+\cos x} = \text{진동}
 \end{aligned}$$

$$\text{따라서 로피탈 정리를 쓸 수 없다.}$$

$$\lim_{x \rightarrow \infty} \frac{\int_0^x \sqrt{1+\cos t} dt}{x}$$

$$= \lim_{x \rightarrow \infty} \frac{\int_0^x \sqrt{2 \frac{1 + \cos t}{2}} dt}{x}$$

$$= \sqrt{2} \lim_{x \rightarrow \infty} \frac{\int_0^x \sqrt{\cos^2 \frac{t}{2}} dt}{x}$$

$$= \sqrt{2} \lim_{x \rightarrow \infty} \frac{\int_0^x \left| \cos \frac{t}{2} \right| dt}{x}$$

$$= 2\sqrt{2} \lim_{x \rightarrow \infty} \frac{\int_0^{\frac{x}{2}} |\cos u| du}{x} \quad (\because \frac{t}{2} = u)$$

여기서 $x = n\pi$ (n 은 자연수)로 치환하면

$$2\sqrt{2} \lim_{x \rightarrow \infty} \frac{\int_0^{\frac{x}{2}} |\cos u| du}{x}$$

$$= 2\sqrt{2} \lim_{n \rightarrow \infty} \frac{\int_0^{\frac{n\pi}{2}} |\cos u| du}{n\pi}$$

$$= 2\sqrt{2} \lim_{n \rightarrow \infty} \frac{n}{n\pi} = \frac{2\sqrt{2}}{\pi}$$

정답 : ②

9) 미분가능한 두 함수 f, g 이고 $f(g(x)) = x$ 이므로

$f^{-1}(x) = g(x)$ 이다.

또한 $f'(x) = 4 + (f(x))^2 = 4 + y^2$ 에서

$g'(y) = \frac{1}{f'(x)} = \frac{1}{4 + y^2}$ 이고 $g(y) = \frac{1}{2} \tan^{-1} \frac{y}{2} + C$ 이다.

$f(0) = 0$ 이므로 $g(0) = 0$ 이고 $C = 0$ 이므로

$g(x) = \frac{1}{2} \tan^{-1} \frac{x}{2}$ 이다. $g(2\sqrt{3}) = \frac{\pi}{6}$ 이므로 $f\left(\frac{\pi}{6}\right) = 2\sqrt{3}$ 이다.

정답 : ①

10)

$\frac{f(x)}{x^3(x-1)^2} = \frac{A}{x} + \frac{B}{x^2} + \frac{C}{x^3} + \frac{D}{x-1} + \frac{E}{(x-1)^2}$ 에서

부정적분이 유리함수가 되어야 하므로 $A = D = 0$ 이다.

따라서 $\frac{B}{x^2} + \frac{C}{x^3} + \frac{E}{(x-1)^2} = \frac{Bx(x-1)^2 + C(x-1)^2 + Ex^3}{x^3(x-1)^2}$ 이므로

$f(x) = Bx(x-1)^2 + C(x-1)^2 + Ex^3$ 이다. 여기서 $f(x)$ 가 이차함수이므로 $B + E = 0$, $f(0) = 1$ 이므로 $C = 1$, $f'(0) = 0$ 이므로

$B - 2C = 0$ 이다. 연립하면 $B = 2$ 이다. 따라서 $f(1) = -2$ 이다.

정답 : ②

$$11) \int_0^1 (\tan^{-1} x)^2 dx - \int_0^1 (\cot^{-1} x)^2 dx$$

$$= \int_0^1 (\tan^{-1} x)^2 - (\cot^{-1} x)^2 dx$$

$$= \int_0^1 (\tan^{-1} x + \cot^{-1} x)(\tan^{-1} x - \cot^{-1} x) dx$$

$$= \frac{\pi}{2} \int_0^1 \left(2\tan^{-1} x - \frac{\pi}{2} \right) dx$$

$$= \frac{\pi}{2} \left[2x \tan^{-1} x - \ln(1+x^2) - \frac{\pi}{2} x \right]_0^1$$

$$= -\frac{\pi}{2} \ln 2$$

정답 : ③

12)

(i) $f(x) \int_0^x f(t) dt = e^x + 3x - 1$ 의 양변을 x 에 대하여 미분하면

$$f'(x) \int_0^x f(t) dt + f(x)^2 = e^x + 3$$

$x = 0$ 을 대입하면 $f(0)^2 = 4$

(ii) $f(x) \int_0^x f(t) dt = e^x + 3x - 1$ 에서 양변을 적분하면

$$\frac{1}{2} \left(\int_0^x f(t) dt \right)^2 = e^x + \frac{3}{2} x^2 - x + C$$

$x = 0$ 일 때 $0 = 1 + C \Rightarrow C = -1$ 이므로

$$\left(\int_0^x f(t) dt \right)^2 = 2 \left(e^x + \frac{3}{2} x^2 - x - 1 \right)$$

$x = 2$ 를 대입하면

$$\left(\int_0^2 f(x) dx \right)^2 = 2 \left(e^2 + \frac{3}{2} (2)^2 - 2 - 1 \right) = 2e^2 + 6$$

(i), (ii)에 의하여

$$f(0)^2 + \left\{ \int_0^2 f(x) dx \right\}^2 = 4 + 2e^2 + 6 = 2e^2 + 10$$

정답 : ③

13)

$$\int_1^e \frac{\ln x^2}{(1 + \ln x)^2} dx$$

$$= 2 \int_1^e \frac{\ln x}{(1 + \ln x)^2} dx \quad (\ln x + 1 = t)$$

$$= 2 \int_1^2 \frac{(t-1)}{t^2} e^{t-1} dt$$

$$= \frac{2}{e} \int_1^2 \left(\frac{1}{t} - \frac{1}{t^2} \right) e^t dt$$

$$= \frac{2}{e} \left[\frac{1}{t} e^t \right]_1^2 \quad (\because \text{곱 미분})$$

$$= \frac{2}{e} \left(\frac{e^2}{2} - e \right) = e - 2$$

정답 : ③

14)

$$\int_{-\sqrt{3}}^{-1} 2 \tan^{-1} x dx + \int_{-\sqrt{3}}^{-1} \sin^{-1} \left(\frac{2x}{1+x^2} \right) dx$$

$\tan^{-1} x = \theta \Leftrightarrow \tan \theta = x$ 라 하면

$$\frac{2x}{1+x^2} = 2 \sin \theta \cos \theta = \sin 2\theta = \sin(2 \tan^{-1} x) \text{이다.}$$

$$\int_{-\sqrt{3}}^{-1} \sin^{-1} \left(\frac{2x}{1+x^2} \right) dx$$

$$= \int_{-\sqrt{3}}^{-1} \sin^{-1}(\sin(2 \tan^{-1} x)) dx \text{이고}$$

$$-\sqrt{3} \leq x \leq -1$$

$$\Rightarrow -\frac{\pi}{3} \leq \tan^{-1} x \leq -\frac{\pi}{4}$$

$$\Rightarrow -\frac{2\pi}{3} \leq 2 \tan^{-1} x \leq -\frac{\pi}{2}$$

$$\Rightarrow \frac{\pi}{3} \leq \pi + 2 \tan^{-1} x \leq \frac{\pi}{2} \text{이므로}$$

$$\int_{-\sqrt{3}}^{-1} \sin^{-1}(\sin(-\pi + \pi + 2 \tan^{-1} x)) dx$$

$$= \int_{-\sqrt{3}}^{-1} \sin^{-1}(-\sin(\pi + 2 \tan^{-1} x)) dx$$

$$= - \int_{-\sqrt{3}}^{-1} \sin^{-1}(\sin(\pi + 2 \tan^{-1} x)) dx$$

$$= - \int_{-\sqrt{3}}^{-1} \pi + 2 \tan^{-1} x dx \text{ 이다.}$$

$$\begin{aligned} \text{따라서 } & \int_{-\sqrt{3}}^{-1} 2 \tan^{-1} x dx + \int_{-\sqrt{3}}^{-1} \sin^{-1} \left(\frac{2x}{1+x^2} \right) dx \\ &= \int_{-\sqrt{3}}^{-1} 2 \tan^{-1} x dx - \int_{-\sqrt{3}}^{-1} \pi + 2 \tan^{-1} x dx \\ &= - \int_{-\sqrt{3}}^{-1} \pi dx = \pi(1 - \sqrt{3}) \end{aligned}$$

정답 : ①

15)

$$x^3 + y^3 = (x+y)^3 - 3xy(x+y) \text{ 이므로}$$

$$\begin{aligned} (\sin^2 x)^3 + (\cos^2 x)^3 &= (\sin^2 x + \cos^2 x)^3 - 3 \sin^2 x \cos^2 x (\sin^2 x + \cos^2 x) \\ &= 1 - 3 \sin^2 x \cos^2 x \text{ 이다.} \end{aligned}$$

$$\int_0^{\frac{\pi}{4}} \frac{1}{\sin^6 x + \cos^6 x} dx$$

$$= \int_0^{\frac{\pi}{4}} \frac{1}{1 - 3 \sin^2 x \cos^2 x} dx$$

$$= \int_0^{\frac{\pi}{4}} \frac{1}{1 - \frac{3}{4} (2 \sin x \cos x)^2} dx$$

$$= \int_0^{\frac{\pi}{4}} \frac{1}{1 - \frac{3}{4} (\sin 2x)^2} dx$$

$$= \int_0^{\frac{\pi}{4}} \frac{4}{4 - 3 \sin^2 2x} dx$$

$$= \int_0^{\frac{\pi}{4}} \frac{4}{4 - 3 \frac{1 - \cos 4x}{2}} dx$$

$$= \int_0^{\frac{\pi}{4}} \frac{8}{5 + 3 \cos 4x} dx \quad (4x = t)$$

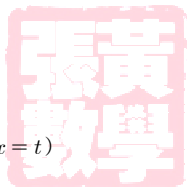
$$= \int_0^{\pi} \frac{2}{5 + 3 \cos t} dt \quad \left(\tan \frac{t}{2} = u \right)$$

$$= \int_0^{\infty} \frac{2}{5 + 3 \frac{1-t^2}{1+t^2}} \frac{2}{1+t^2} dt$$

$$= \int_0^{\infty} \frac{2}{4+t^2} dt$$

$$= 2 \left[\frac{1}{2} \tan^{-1} \left(\frac{t}{2} \right) \right]_0^{\infty} = \frac{\pi}{2}$$

정답 : ④



장황수학